

VPC USING BASTION SERVER

1. VPC with Bastion Server

VPC & Subnet Design

VPC CIDR: 10.0.0.0/16

Public Subnet (Bastion Server): 10.0.1.0/24

Private Subnet (Database Server): 10.0.2.0/24

2. Security Group Configuration (VERY IMPORTANT)

Bastion Server Security Group (Public Subnet)

Allows SSH (22) from: My IP (Database Administrator)

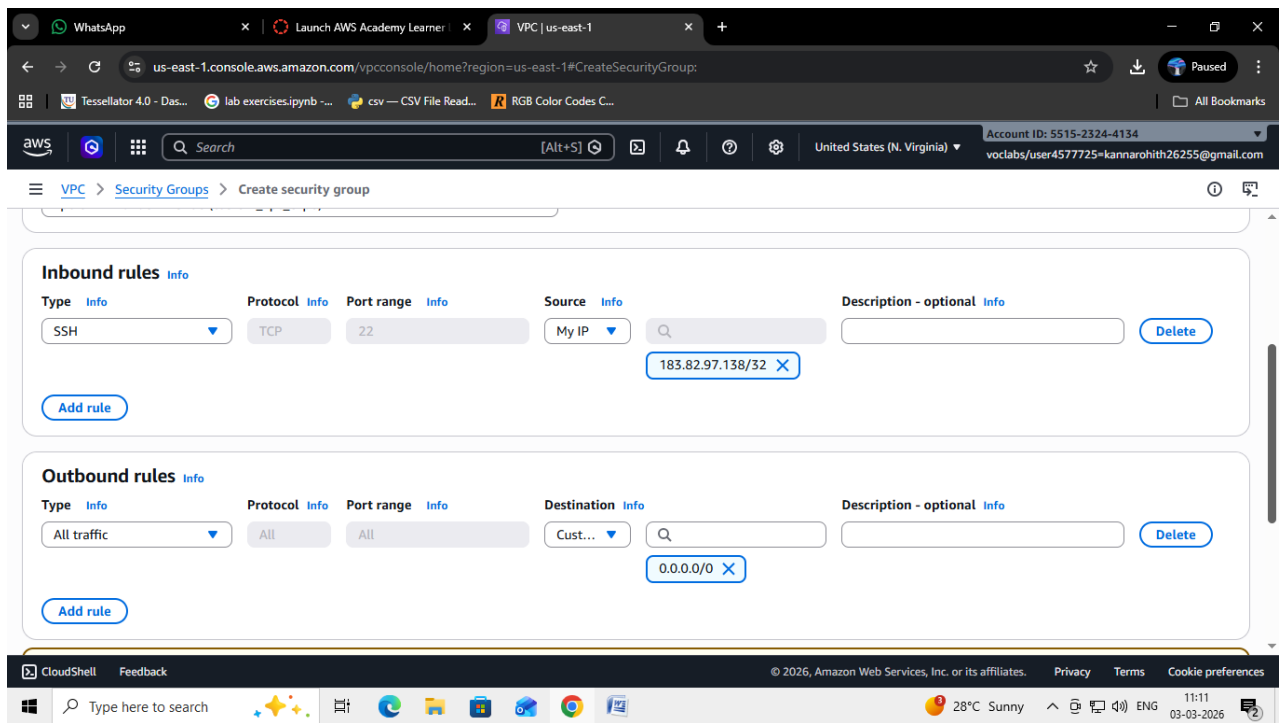
The screenshot displays the AWS Management Console interface for creating a new security group. The browser address bar shows the URL: `us-east-1.console.aws.amazon.com/vpconsole/home?region=us-east-1#CreateSecurityGroup:`. The console header indicates the user is logged in as `voclabs/user4577725-kannarohith26255@gmail.com` in the `us-east-1` region.

The main content area is titled **Create security group** and includes a brief description: "A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below."

The form is divided into two main sections:

- Basic details**:
 - Security group name**: `BastionSG`. A note states: "Name cannot be edited after creation."
 - Description**: `Allows SSH`.
 - VPC**: `vpc-0727f4d7d617191a6 (custom_vpc_exp6)`.
- Inbound rules**:
 - Type**: `SSH`.
 - Protocol**: `TCP`.
 - Port range**: `22`.
 - Source**: `My IP`.
 - Description - optional**: (Empty field).
 - Delete**: (Button).

The bottom of the screenshot shows the Windows taskbar with the system clock indicating 11:11 on 03-03-2026, and the weather as 28°C Sunny.

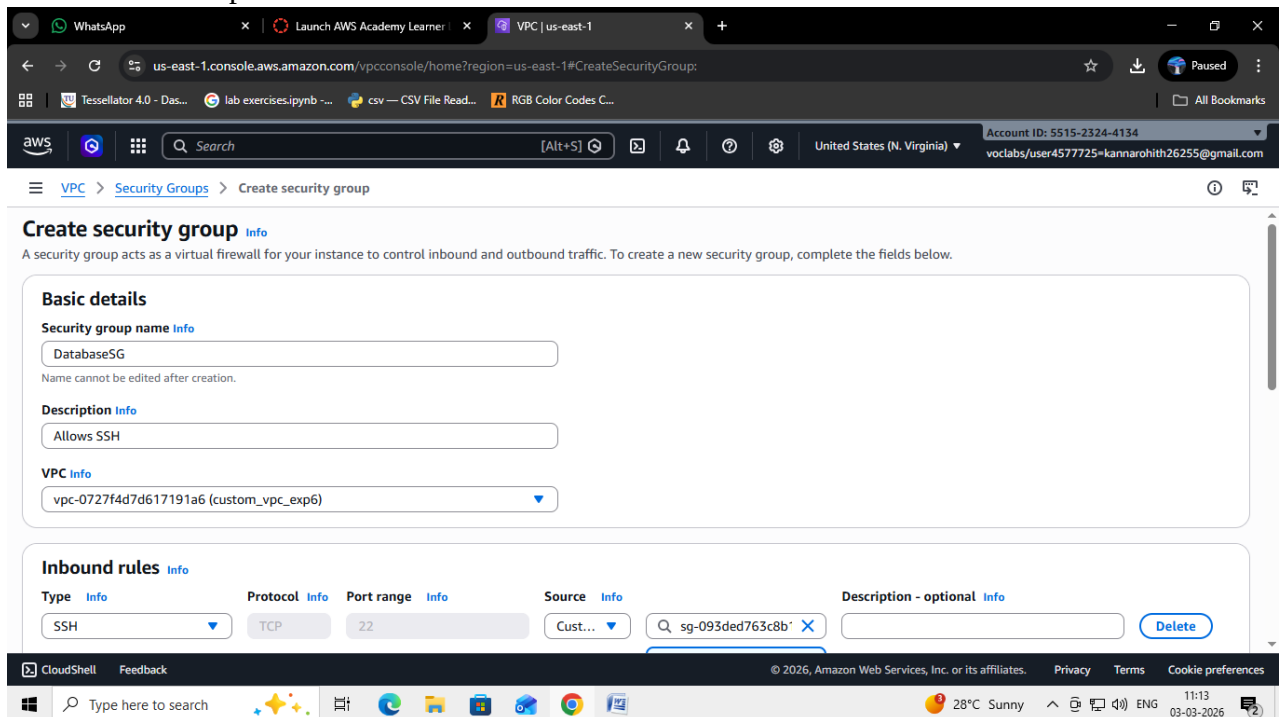


Database Server Security Group (Private Subnet)

Allows MySQL / Aurora (3306) from: custom 10.0.1.0/24 (Public subnet / Bastion subnet)

Allows SSH (22) from: custom Private IP of Bastion Server

Does NOT allow public access



The screenshot shows the AWS Management Console interface for creating a security group. The 'Inbound rules' section contains two rules: one for SSH (Type: SSH, Protocol: TCP, Port range: 22, Source: Custom, Destination: sg-093ded763c8b117c8) and one for MySQL/Aurora (Type: MySQL/Aurora, Protocol: TCP, Port range: 3306, Source: Custom, Destination: 10.0.1.0/24). The 'Outbound rules' section contains one rule for All traffic (Type: All traffic, Protocol: All, Port range: All, Destination: Custom, Destination: 0.0.0.0/0). A warning message states: 'Rules with destination of 0.0.0.0/0 or :::0 allow your instances to send traffic to any IPv4 or IPv6 address. We recommend setting security group rules to be more restrictive and to only allow traffic to specific known IP addresses.' The 'Tags - optional' section is empty. The 'Create security group' button is visible at the bottom right.

3. Security Outcome

Database server is NOT publicly accessible
 Only Bastion server can connect to DB server using SSH
 External users cannot access DB directly

4. EC2 Instance Creation

Bastion Server

Subnet: Public Subnet (10.0.1.0/24)
 Key pair: bastion.pem
 Operating System: Ubuntu

EC2 > Instances > Launch an instance

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags Info

Name

public

Add additional tags

Application and OS Images (Amazon Machine Image) Info

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Search our full catalog including 1000s of application and OS images

Recents

Quick Start

Amazon Linux

macOS

Ubuntu

Windows

Red Hat

SUSE Linux

Del

Browse more AMIs

Summary

Number of instances Info

1

Software Image (AMI)

Canonical, Ubuntu, 24.04, amd64... [read more](#)
ami-0b6c6ebd2801a5cb

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

Cancel

Launch instance

Preview code

EC2 > Instances > Launch an instance

On-Demand Linux base pricing: 0.0104 USD per Hour
On-Demand Windows base pricing: 0.0196 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0139 USD per Hour
On-Demand SUSE base pricing: 0.0104 USD per Hour
On-Demand RHEL base pricing: 0.0392 USD per Hour

Additional costs apply for AMIs with pre-installed software

All generations

Compare instance types

Key pair (login) Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

bastion

Create new key pair

Network settings Info

Network Info

vpc-08e6f3ebe3f57f6d6

Edit

Summary

Number of instances Info

1

Software Image (AMI)

Canonical, Ubuntu, 24.04, amd64... [read more](#)
ami-0b6c6ebd2801a5cb

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

Cancel

Launch instance

Preview code

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#LaunchInstances:

EC2 > Instances > Launch an instance

VPC - required | Info

vpc-0727f4d7d617191a6 (custom_vpc_exp6)
10.0.0.0/16

Subnet | Info

subnet-0693708a84a07ffa public_subnet
VPC: vpc-0727f4d7d617191a6 Owner: 551523244134
Availability Zone: us-east-1a (use1-az6) Zone type: Availability Zone
IP addresses available: 251 CIDR: 10.0.1.0/24

Auto-assign public IP | Info

Enable

Firewall (security groups) | Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☐ Create security group ☒ Select existing security group

Common security groups | Info

Select security groups

BastionSG sg-093ded763c8b117c8
VPC: vpc-0727f4d7d617191a6

Summary

Number of instances | Info

1

Software Image (AMI)

Canonical, Ubuntu, 24.04, amd64...read more
ami-0b6c6ebd2801a5cb

Virtual server type (instance type)

t3.micro

Firewall (security group)

BastionSG

Storage (volumes)

Cancel Launch instance Preview code

Database Server

Subnet: Private Subnet (10.0.2.0/24)

Key pair: dbserver.pem

Operating System: Ubuntu

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#LaunchInstances:

EC2 > Instances > Launch an instance

Name and tags | Info

Name

private Add additional tags

Application and OS Images (Amazon Machine Image) | Info

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Search our full catalog including 1000s of application and OS images

Recents Quick Start

Amazon Linux macOS Ubuntu Windows Red Hat SUSE Linux Del

aws ubuntu Microsoft Red Hat SUSE

Browse more AMIs
Including AMIs from AWS, Marketplace and

Summary

Number of instances | Info

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.10...read more
ami-0f3caa1cf4417e51b

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

Cancel Launch instance Preview code

EC2 > Instances > Launch an instance

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - **required**

dbserver [Create new key pair](#)

Network settings [Info](#) [Edit](#)

Network [Info](#)

vpc-08e6f3ebe3f57f6d6

Subnet [Info](#)

No preference (Default subnet in any availability zone)

Auto-assign public IP [Info](#)

Enable

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group ☐ Select existing security group

Summary

Number of instances [Info](#)

1

Software Image (AMI)

Canonical, Ubuntu, 24.04, amd64...[read more](#)
ami-0b6c6ebd2801a5cb

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

[Cancel](#) [Launch instance](#) [Preview code](#)

EC2 > Instances > Launch an instance

VPC - required [Info](#)

vpc-0727f4d7d617191a6 (custom_vpc_exp6) 10.0.0.0/16 [Create new vpc](#)

Subnet [Info](#)

subnet-0364e47f192a591ec private_subnet [Create new subnet](#)

VPC: vpc-0727f4d7d617191a6 Owner: 551523244134
Availability Zone: us-east-1a (use1-az6) Zone type: Availability Zone
IP addresses available: 251 CIDR: 10.0.2.0/24)

Auto-assign public IP [Info](#)

Disable

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☐ Create security group ☒ Select existing security group

Common security groups [Info](#)

Select security groups [Compare security group rules](#)

DatabaseSG sg-0d1043775e251f9a6 [X](#)
VPC: vpc-0727f4d7d617191a6

Summary

Number of instances [Info](#)

1

Software Image (AMI)

Canonical, Ubuntu, 24.04, amd64...[read more](#)
ami-0b6c6ebd2801a5cb

Virtual server type (instance type)

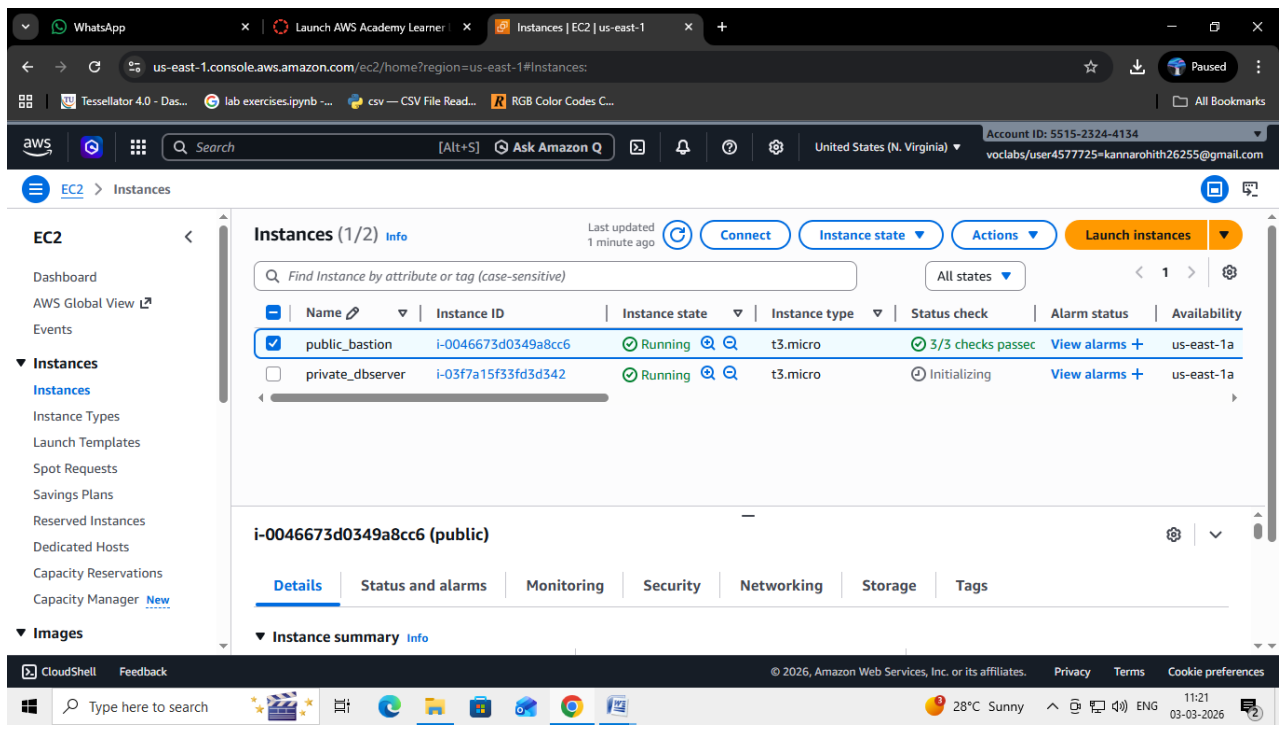
t3.micro

Firewall (security group)

DatabaseSG

Storage (volumes)

[Cancel](#) [Launch instance](#) [Preview code](#)



5. Copy DB Server Key to Bastion Server

Database server is in private subnet

Direct SSH from local system is NOT possible

DB server key must be copied to Bastion server

Command (PowerShell):

```
scp -i bastion.pem dbserver.pem ubuntu@<BASTION_PUBLIC_IP>:/home/ubuntu/
```

File is copied securely using SCP

dbserver.pem will now be available inside Bastion server

```
fs@DESKTOP-GGREQK6 MINGW64 ~/Desktop/aws (master)
$ scp -i bastion.pem dbserver.pem ubuntu@107.23.32.234:/home/ubuntu
dbserver.pem
```

6. Connect to Bastion Server

SSH Command:

```
ssh -i bastion.pem ubuntu@<BASTION_PUBLIC_IP>
```

Verify key files:

```
ls
```

Output:

```
bastion.pem
```

```
dbserver.pem
```

```
Connection to 107.23.32.234 closed.

fs@DESKTOP-GGREQK6 MINGW64 ~/Desktop/aws (master)
$ scp -i bastion.pem dbserver.pem ubuntu@107.23.32.234:/home/ubuntu
dbserver.pem

fs@DESKTOP-GGREQK6 MINGW64 ~/Desktop/aws (master)
$ ssh -i "bastion.pem" ubuntu@107.23.32.234
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1018-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Tue Mar  3 06:26:53 UTC 2026

System load:  0.0               Temperature:   -273.1 C
Usage of /:   26.4% of 6.71GB   Processes:    113
Memory usage: 24%              Users logged in: 0
Swap usage:   0%               IPv4 address for ens5: 10.0.1.175

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

Last login: Tue Mar  3 06:10:55 2026 from 183.82.97.138
ubuntu@ip-10-0-1-175:~$ ls
dbserver.pem
ubuntu@ip-10-0-1-175:~$
```

7. Connect from Bastion to Database Server

SSH from Bastion:

```
ssh -i dbserver.pem ubuntu@10.0.2.x
```

Connection to DB server is successful

Database administrator can now:

Update database configurations

Install required packages-----problem occurs

Restart MySQL service

```
ubuntu@ip-10-0-1-175:~$ ls
dbserver.pem
ubuntu@ip-10-0-1-175:~$ ch mod 400 dbserver.pem
ch: command not found
ubuntu@ip-10-0-1-175:~$ chmod 400 dbserver.pem
ubuntu@ip-10-0-1-175:~$ ssh -i "dbserver.pem" ubuntu@10.0.2.135
The authenticity of host '10.0.2.135 (10.0.2.135)' can't be established.
ED25519 key fingerprint is SHA256:8sbYhPp+IaecCDE0sAsNTn9ayGmpCVk/Rw0Jj7CCJhQ.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.0.2.135' (ED25519) to the list of known hosts.
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1018-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Tue Mar  3 06:34:06 UTC 2026

System load:  0.0           Temperature:   -273.1 C
Usage of /:   26.4% of 6.71GB Processes:    112
Memory usage: 25%          Users logged in: 0
Swap usage:   0%           IPv4 address for ens5: 10.0.2.135

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
```


us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#Addresses:PublicIp=107.22.78.239

Account ID: 5515-2324-4134
voclabs/user4577725-kannarohith26255@gmail.com

VPC > Elastic IP addresses

Filter by VPC

Virtual private cloud

- Your VPCs
- Subnets
- Route tables
- Internet gateways
- Egress-only internet gateways
- Carrier gateways
- DHCP option sets
- Elastic IPs**
- Managed prefix lists
- NAT gateways
- Peering connections
- Route servers

Security

- Network ACLs

Elastic IP address allocated successfully.
Elastic IP address 107.22.78.239

Associate this Elastic IP address

Elastic IP addresses (1) Info

Find elastic IP addresses by attribute or tag

Public IPv4 address : 107.22.78.239

Clear filters

☐	Name	Allocated IPv4 address	Type	Allocation ID	Reverse DNS
☐	-	107.22.78.239	Public IP	eipalloc-0680757e4048a6bc8	-

Select an elastic IP address

View IP address usage and recommendations to release unused IPs with [Public IP insights](#)

CloudShell Feedback

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Type here to search

India reported 4.8%...

12:15
03-03-2026

NAT GATEWAY:

us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#CreateNatGateway:

Account ID: 5515-2324-4134
voclabs/user4577725-kannarohith26255@gmail.com

VPC > NAT gateways > Create NAT gateway

Create NAT gateway Info

A highly available, managed Network Address Translation (NAT) service that instances in private subnets can use to connect to services in other VPCs, on-premises networks, or the internet.

NAT gateway settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.

nat-gw

The name can be up to 256 characters long.

Availability mode Info
Choose whether to deploy across all zones in the region or restrict to a single availability zone.

☒ **Regional - new**
Scales automatically across all regional AZs, simplifying management for multi AZ deployments.

☐ **Zonal**
Provides granular control within a specific availability zone, adhering to subnet level settings.

VPC
Select a VPC in which to create the regional NAT gateway.

vpc-0727f4d7d617191a6 (custom_vpc_exp6)

Connectivity type
Select a connectivity type for the NAT gateway.

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Type here to search

Air Poor

12:16
03-03-2026

us-east-1.console.aws.amazon.com/vpconsole/home?region=us-east-1#CreateNatGateway:

VPC > NAT gateways > Create NAT gateway

VPC
Select a VPC in which to create the regional NAT gateway.
vpc-0727f4d7d617191a6 (custom_vpc_exp6)

Connectivity type
Select a connectivity type for the NAT gateway.
☒ Public
☐ Private

Method of Elastic IP (EIP) allocation Info
Choose how IP addresses are associated with NAT gateways.
☒ Automatic
AWS automatically manages EIPs and AZ coverage for NAT gateways. This ensures easy scaling-- adding AZs automatically allocates EIPs, simplifying management.
☐ Manual
Manually assigns specific IP addresses for compliance or whitelisting. Note: Requires manual scaling to new AZs as workloads expand.

Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.
Key Value - optional

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us-east-1.console.aws.amazon.com/vpconsole/home?region=us-east-1#NatGatewayDetails:natGatewayId=nat-16352a1bb856929b1

VPC > NAT gateways > nat-16352a1bb856929b1

Filter by VPC

Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only internet gateways
Carrier gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Route servers

Security
Network ACLs

NAT gateway nat-16352a1bb856929b1 | nat-gw was created successfully.

nat-16352a1bb856929b1 / nat-gw Actions

Details

NAT gateway ID nat-16352a1bb856929b1	Availability mode Regional	State Pending	State message Info -
NAT gateway ARN arn:aws:ec2:us-east-1:55152324:4134:natgateway/nat-16352a1bb856929b1	Connectivity type Public	Created Tuesday, March 3, 2026 at 12:17:10 GMT+5:30	Deleted -
VPC vpc-0727f4d7d617191a6 / custom_vpc_exp6	Method of EIP allocation Automatic		

IP addresses Monitoring Flow logs Tags

Associated IP addresses Edit IP address associations

Route_table:

Create route table

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

Route table settings

Name - optional

Create a tag with a key of 'Name' and a value that you specify.

private_route

VPC

The VPC to use for this route table.

vpc-0727f4d7d617191a6 (custom_vpc_exp6)

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Name

Value - optional

private_route

Remove

Add new tag

You can add 49 more tags.

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (1/2)

Filter subnet associations

	Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
☐	public_subnet	subnet-0693708a84a07fffa	10.0.1.0/24	-	rtb-03d3c8ce882301183 / public_rout..
☒	private_subnet	subnet-0364e47f192a591ec	10.0.2.0/24	-	Main (rtb-0faa608f3d7a42d89)

Selected subnets

subnet-0364e47f192a591ec / private_subnet

Cancel

Save associations

us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#EditRoutes:RouteTableId=rtb-0d52b45dfba29bb2d

Account ID: 5515-2324-4134
voclabs/user4577725-kannarohith26255@gmail.com

VPC > Route tables > rtb-0d52b45dfba29bb2d > Edit routes

Edit routes

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	CreateRouteTable
0.0.0.0/0	NAT Gateway	-	No	CreateRoute

Remove

Add route

Cancel Preview Save changes

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Type here to search High UV 12:19 03-03-2026

In gitbash for database server access by nat gateway:

```
W: Failed to fetch http://security.ubuntu.com/ubuntu/dists/noble-security/InRelease Cannot initia
c80:0:1000::17). - connect (101: Network is unreachable) Cannot initiate the connection to securi
101: Network is unreachable) Cannot initiate the connection to security.ubuntu.com:80 (2620:2d:40
Cannot initiate the connection to security.ubuntu.com:80 (2620:2d:4002:1::103). - connect (101: Ne
n to security.ubuntu.com:80 (2620:2d:4000:1::103). - connect (101: Network is unreachable) Cannot
2620:2d:4000:1::101). - connect (101: Network is unreachable) Cannot initiate the connection to se
ct (101: Network is unreachable) Cannot initiate the connection to security.ubuntu.com:80 (2620:2d
le) Cannot initiate the connection to security.ubuntu.com:80 (2a06:bc80:0:1000::16). - connect (10
urity.ubuntu.com:80 (91.189.91.82), connection timed out Could not connect to security.ubuntu.com
connect to security.ubuntu.com:80 (185.125.190.83), connection timed out Could not connect to se
d out Could not connect to security.ubuntu.com:80 (91.189.92.22), connection timed out Could not
nnection timed out Could not connect to security.ubuntu.com:80 (185.125.190.82), connection timed
.189.91.83), connection timed out Could not connect to security.ubuntu.com:80 (91.189.92.24), con
W: Some index files failed to download. They have been ignored, or old ones used instead.
ubuntu@ip-10-0-2-135:~$ sudo apt update
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:3 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1494 kB]
Get:4 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [240 kB]
Get:5 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.6 kB]
Get:6 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Metadata [9932 B]
Get:7 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [973 kB]
Get:8 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [218 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [74.3 kB]
Get:10 http://security.ubuntu.com/ubuntu noble-security/universe amd64 c-n-f Metadata [20.6 kB]
Get:11 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [2568 kB]
Get:12 http://security.ubuntu.com/ubuntu noble-security/restricted Translation-en [596 kB]
Get:13 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [212 B]
Get:14 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 c-n-f Metadata [536 B]
Get:15 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Packages [28.8 kB]
Get:16 http://security.ubuntu.com/ubuntu noble-security/multiverse Translation-en [6732 B]
Get:17 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [208 B]
Get:18 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 c-n-f Metadata [396 B]
Get:19 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:20 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:21 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 Packages [15.0 MB]
Get:22 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe Translation-en [5982 kB]
Get:23 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 Components [3871 kB]
```



```
ubuntu@ip-10-0-2-135: ~$
Get:22 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe Translation-en [5982 kB]
Get:23 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 Components [3871 kB]
Get:24 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 c-n-f Metadata [301 kB]
Get:25 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 Packages [269 kB]
Get:26 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse Translation-en [118 kB]
Get:27 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 Components [135.0 kB]
Get:28 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 c-n-f Metadata [8328 B]
Get:29 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1800 kB]
Get:30 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main Translation-en [331 kB]
Get:31 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [175 kB]
Get:32 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [16.5 kB]
Get:33 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1558 kB]
Get:34 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [316 kB]
Get:35 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [385 kB]
Get:36 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 c-n-f Metadata [32.7 kB]
Get:37 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [2735 kB]
Get:38 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [630 kB]
Get:39 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:40 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 c-n-f Metadata [556 B]
Get:41 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Packages [32.1 kB]
Get:42 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/multiverse Translation-en [7044 B]
Get:43 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:44 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 c-n-f Metadata [496 B]
Get:45 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/main amd64 Packages [40.4 kB]
Get:46 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/main Translation-en [9208 B]
Get:47 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7284 B]
Get:48 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/main amd64 c-n-f Metadata [368 B]
Get:49 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/universe amd64 Packages [29.5 kB]
Get:50 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/universe Translation-en [17.9 kB]
Get:51 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [10.5 kB]
Get:52 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/universe amd64 c-n-f Metadata [1444 B]
Get:53 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [216 B]
Get:54 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/restricted amd64 c-n-f Metadata [116 B]
Get:55 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [212 B]
Get:56 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 c-n-f Metadata [116 B]
Fetched 40.4 MB in 7s (5894 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
99 packages can be upgraded. Run 'apt list --upgradable' to see them.
ubuntu@ip-10-0-2-135:~$
```

SCENARIO BASED QUESTIONS

1. You have multiple private subnets in different AZs. Should you use **one NAT Gateway** or **multiple NAT Gateways**? Explain your design choice.
2. Your private subnet instances can reach the internet, but the NAT Gateway shows no traffic. What could be the possible reasons?
3. A NAT Gateway was accidentally created in a **private subnet** instead of a public subnet. What problem will occur, and how would you fix it?
4. An application running in a private subnet requires outbound access only to a few trusted external services. How would you restrict traffic while still using a NAT Gateway?
5. Multiple teams share a single VPC and NAT Gateway. One team's traffic is affecting others. How would you redesign the NAT Gateway setup to isolate traffic?
6. You can successfully SSH into a private EC2 instance using a Bastion server, but yum update / apt update fails inside the private instance. Why?
7. Private EC2 instances can access the internet through a NAT Gateway, but administrators cannot SSH into them using the Bastion server.
8. A Bastion server and NAT Gateway are both created in a **private subnet** by mistake. What impact does this cause?
9. SSH access to the Bastion server works, but SSH from Bastion to the private EC2 fails. Why?
10. A student claims that a Bastion server alone is enough to provide internet access to private instances. Why is Bastion not a replacement for NAT Gateway?